

SCovox map accuracy vs GLIM full SLAM map

Date: 2026-06-29 **Bag:** 2026_06_19_18_19_06__kalhan-map-test-2_ (Ouster + IMU, ~500 s)

Pipeline: GLIM LiDAR-IMU SLAM → TF → SCovox occupancy, *live*, GLIM-deskewed feed.

Reference: GLIM dense factor-graph map (`glim_map.pcd`) — treated as geometric ground truth.

Configuration under test

param	value
input cloud	<code>/glim_ros/points</code> (GLIM-deskewed, frame <code>imu</code>)
base_frame / integration_frame	<code>imu</code> / <code>odom</code>
resolution	0.10 m
max_range	15.0 m
occupancy_vis_threshold	0.6

Config: `scovox/config/glim/scovox_lidar_glim.yaml` .

Method

1. Both clouds voxel-downsampled at 0.10 m.
2. SCovox (`odom`) rigidly ICP-aligned onto GLIM (`map`), outlier-gated at 1 m so the alignment is not biased by any sparse far voxels.
3. Bidirectional nearest-neighbour distances (scipy `cKDTree`): - **SCovox** → **GLIM** = precision/accuracy (is each SCovox voxel a real surface?) - **GLIM** → **SCovox** = recall/coverage (is each GLIM surface mapped?)
4. **Fair coverage:** recall recomputed over only the GLIM surfaces within `max_range` (15 m) of the trajectory — i.e. the region SCovox actually attempts to map.

Same-run pair. The live SCovox map was salvaged from the running node via

`glim_localisation/scripts/glim/salvage_capture_seq.py` (advancing `/clock` + `SALVAGE_RELIABLE=1`). Comparison tool:

`glim_localisation/scripts/glim/compare_scovox_glim.py` .

Inputs

	raw points	@0.10 m voxels	bbox (m)
SCovox (<code>odom</code>)	1,043,389	575,636	82 × 100 × 18
GLIM (<code>map</code>)	12,250,000	4,270,126	232 × 249 × 56

ICP odom → map : **|t| = 0.095 m, RMSE 0.072 m** → GLIM was essentially drift-free on this bag, so odom ≈ map and the metrics below are alignment-robust.

Results

Accuracy / precision (SCovox → GLIM) — excellent

metric	value
median voxel → surface error	0.057 m (~½ voxel)
mean / RMSE	0.063 / 0.075 m
p95 / p99 / max	0.118 / 0.191 / 4.17 m
precision @0.2 m / @0.3 m	0.991 / 0.998
spurious voxels (> 0.5 m from any GLIM surface)	0.1 %

Nearly every SCovox voxel lies on a real GLIM surface.

Coverage (GLIM → SCovox)

denominator	recall @0.2 m	recall @0.5 m	median gap
full GLIM extent	0.30	0.33	6.4 m
within 15 m of trajectory (fair)	0.72	0.77	0.098 m

Only **41.9 %** of GLIM's full map lies within SCovox's 15 m range gate; the rest is far/tall structure SCovox excludes **by design**. Restricted to the mappable near-field, coverage is 72–77 % with ~10 cm median gap. The ~25 % shortfall is occlusion / observation geometry plus the 0.6 occupancy-probability pruning.

F-score (full-extent, for reference)

τ (m)	precision	recall	F
0.10	0.909	0.216	0.349
0.20	0.991	0.302	0.463
0.30	0.998	0.315	0.479
0.50	0.999	0.328	0.493

Vertical extent / over-fill

	z range (m)	z span	z (p1 ... p99)
GLIM	-7.27 ... 48.38	55.65	-1.24 ... 28.74
SCovox	-1.50 ... 16.30	17.80	-1.10 ... 13.20

0 % of SCovox voxels exceed GLIM's z-envelope. **The historical ~6–7 m vertical over-fill smear (raw- /ouster/points feed) is absent** — the GLIM-deskewed feed is the fix.

Verdict

SCovox produces a **geometrically faithful, low-noise** local occupancy map (~6 cm median accuracy, ~0 % spurious occupancy) that covers ~72–77 % of the near-field (≤ 15 m) GLIM geometry. It is intentionally a tight, planning-scale subset of GLIM's full-extent SLAM map — that is correct behaviour, not a defect.

Reproduce

```
# from HMR_Explo/ws/src
./run_glim_experiment.sh map "" 0.5          # builds glim_map.pcd + scovox_map.npy
# then, in the glim container:
python3 /ws/scripts/glim/compare_scovox_glim.py \
    /ws/output/scovox_node.npy /ws/output/glim_map.pcd 0.10 /ws/output/path_glim.csv
```

Artifacts: `glim_localisation/output/{glim_map.pcd, scovox_node.npy, path_glim.csv, scovox_vs_glim_eval.md}`.